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couplr: Optimal pairing and matching via linear assignment

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Summary

`couplr` (Colling, 2026) is an R package for optimal pairing, matching, and assignment. It solves linear assignment problems: given two sets of objects and a cost for each possible pair, choose the pairs with minimum total cost. That operation appears in causal inference, experimental design, task allocation, sample pairing, and image alignment.

The package provides two entry points. A high-level data-frame interface prepares data, builds distances, enforces constraints, and returns analysis-ready matched output for optimal, greedy, propensity-score, full, coarsened-exact, subclassification, and cardinality matching, with balance diagnostics and Rosenbaum sensitivity bounds (Rosenbaum, 2002). A low-level interface exposes the solvers directly when the user already has a cost matrix. An automatic dispatcher routes each problem to one of 19 internal solvers based on shape, sparsity, cost type, and size, so the same package supports statistical matching workflows and general-purpose assignment tasks.

Statement of need

A serious matching workflow in R today threads three packages by hand: one for the assignment solver, one for preprocessing and constraint construction, one for balance diagnostics. Any non-default choice (exact blocking, calipers on multiple variables, pairwise distances on observed covariates, variable-ratio full matching, replacement matching, direct control over the solver) pulls the user into a different corner of that toolchain.

`couplr` packages the workflow around the linear assignment problem. A causal-inference user starts with a data frame, observed covariates, and a treatment indicator; an operations-research user starts with a cost matrix already in hand. The same solver core serves both.

State of the field

`MatchIt` (Ho et al., 2011), `optmatch` (Hansen & Klopfer, 2006), `Matching` (Sekhon, 2011), and `designmatch` (Zubizarreta et al., 2024) handle preprocessing and matching for causal inference; `cobalt` supplies cross-package balance diagnostics (Greifer, 2025). For the assignment problem itself, R users reach for `clue` (Hungarian via `solve_LSAP`) or `lpSolve` (general LP) (Berkelaar & others, 2024; Hornik, 2024). The causal-inference packages assume a treated/control framing and route through a single fixed back-end (`optmatch` for `MatchIt`, network flow for `optmatch`, genetic search for `Matching`, integer programming via commercial solvers for `designmatch`); the solver-focused packages expose one algorithm each without the preprocessing, constraint, or diagnostic machinery a matching workflow needs.

`couplr` combines both. It treats covariate distance as the primary entry point rather than a side option behind a propensity score, and ships 19 assignment solvers spanning classical

dense assignment, sparse and rectangular variants, k-best (Murty, 1968), bottleneck, min-cost flow, and entropy-regularized transport, with an automatic dispatcher. To our knowledge, `couplr` provides the first publicly available open-source implementation of the Gabow–Tarjan bit-scaling assignment algorithm (Gabow & Tarjan, 1989) in any language; the algorithm's $O(\sqrt{n}\, m \log(n\, C_{\max}))$ bound has stood since 1989 without a library counterpart.

Software design

The package has three layers. The first validates inputs and converts data frames into cost matrices, handling scaling, missing-data checks, and user weights. The second solves the linear assignment problem: given a cost matrix $C \in \mathbb{R}^{n \times m}$ with entries C_{ij} for row $i \in \{1, \dots, n\}$ and column $j \in \{1, \dots, m\}$, find a binary assignment matrix $X \in \{0, 1\}^{n \times m}$ ($X_{ij} = 1$ iff row i is matched to column j) solving

$$\min_X \sum_{i,j} C_{ij} X_{ij} \quad \text{subject to} \quad \sum_j X_{ij} \le 1 \ \forall i, \quad \sum_i X_{ij} \le 1 \ \forall j.$$

The bipartite structure makes the LP relaxation integral (Birkhoff–von Neumann), so row potentials $u_i \in \mathbb{R}$ ($i = 1, \dots, n$) and column potentials $v_j \in \mathbb{R}$ ($j = 1, \dots, m$) satisfying

$$u_i + v_j \le C_{ij} \quad \forall (i, j), \qquad u_i + v_j = C_{ij} \quad \text{whenever } X_{ij} = 1$$

certify optimality; `assignment_duals()` returns (u, v). Worst-case time is $O(n^3)$ for Hungarian and Jonker–Volgenant, $O(n^3 \log(n\, C_{\max}))$ for cost scaling, and $O(\sqrt{n}\, m \log(n\, C_{\max}))$ for Gabow–Tarjan bit scaling, where $C_{\max} = \max_{i,j} |C_{ij}|$ is the largest cost magnitude; the dispatcher selects among solvers using n, m, sparsity, and rectangularity. All 19 solvers are implemented from scratch in C++ via Rcpp, so the package adds no external solver dependency: the Hungarian method (Kuhn, 1955), Jonker–Volgenant shortest augmenting paths (Jonker & Volgenant, 1987), auction algorithms (Bertsekas, 1988), cost scaling (Goldberg & Kennedy, 1995), Gabow–Tarjan scaling (Gabow & Tarjan, 1989), push-relabel min-cost flow ideas (Goldberg & Tarjan, 1988), network simplex, Ramshaw–Tarjan rectangular assignment (Ramshaw & Tarjan, 2012), and related specialized solvers. Each optimal solver is checked against brute-force enumeration over randomised integer, fractional, rectangular, maximizing, and forbidden-edge instances, so the shipped path and the verified path are the same implementation. The third layer returns structured result objects with pairs, costs, diagnostics, weights, subclass information, and conversion methods for other R matching tools.

Distances cover Euclidean, Manhattan, Mahalanobis (pooled within-group covariance by default), Chebyshev, and squared-Euclidean metrics, or a user-supplied function; propensity-score distances use a separate entry point. Three constraint mechanisms reshape the cost matrix before it reaches the solver: a global `max_distance` ceiling, per-variable calipers, and forbidden pairs marked as non-finite entries, which every solver recognises as a missing edge and uses to short-circuit on infeasibility. Blocking either reuses a factor variable or constructs blocks via k-means or hierarchical clustering, and the solver is called once per block.

The dispatcher in `lap_solve(method = "auto")` chooses by shape, sparsity, cost type, and size: dense problems use Jonker–Volgenant; matrices over half forbidden use its sparse variant `lapmod`; strongly rectangular problems avoid padding through shortest augmenting paths; binary and very small problems use specialized solvers. Other solvers are named explicitly. Figure 1 shows median solve time across problem sizes for all 19 solvers.

Cost storage is a second automatic choice. Under `memory_mode = "lazy"` each cost is evaluated from the underlying feature data as the solver requests it, so the footprint follows the covariates rather than the pairs: 100,000 units on 100 covariates hold as 80 MB of features where the dense matrix needs 80 GB. Lazy evaluation covers Jonker–Volgenant and auction with the

built-in metrics, calipers, and `max_distance`, and returns the same assignment as the dense path. The default `"auto"` weighs the estimated dense footprint against free system memory and switches before an oversized allocation.

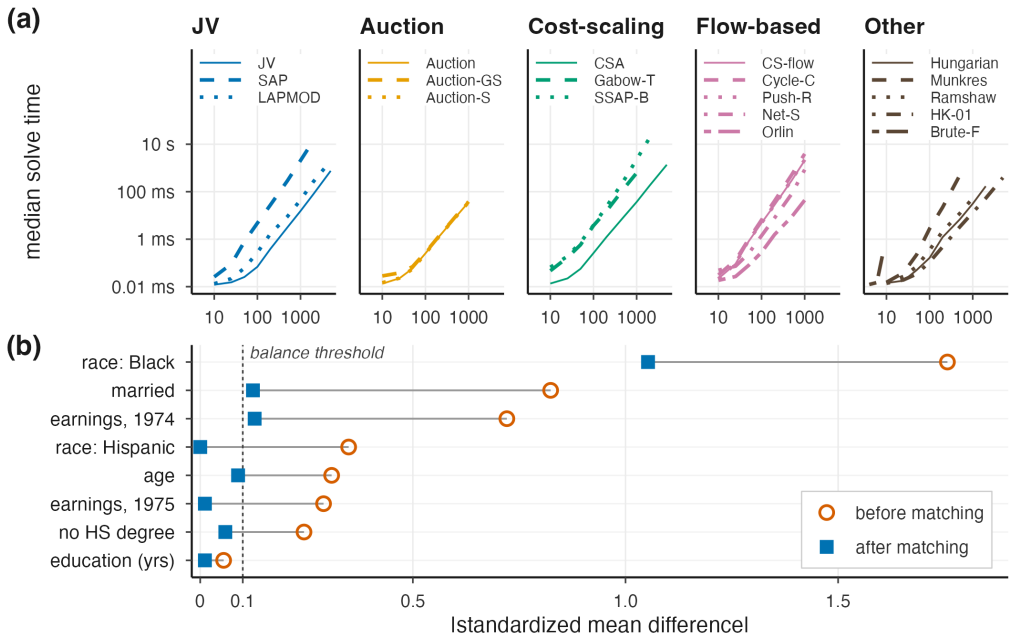

Figure 1: (a) Median wall-clock solve time versus problem size n for all 19 assignment solvers in `couplr`, arranged as five small-multiple panels grouped by algorithm family (JV / augmenting-path, Auction, Cost-scaling, Flow-based, and Other). Within each panel, individual solvers share a single family colour and are distinguished by line style; all panels share log–log axes. Median of 5 replicates on a single core of an Apple M4 Pro; integer cost matrices with entries in $[1, 10{,}000]$. (b) Absolute standardized mean differences on the LaLonde NSW data (185 treated, 429 control, eight covariates) before and after 1-to-1 optimal Mahalanobis matching. After matching, `couplr`, `MatchIt`, and `optmatch` produce identical $|SMD|$ to three decimal places, so a single matched point per covariate is shown. Seven of the eight covariates fall below the 0.1 balance threshold (dashed line); the `race_Black` indicator is reduced from 1.757 to 1.053, a residual no 1-to-1 matcher can resolve here. Reproducible from `paper/make-figure.R` and `paper/bench_lalonde.R`.

`balance_diagnostics()` reports standardized mean differences, variance ratios, and Kolmogorov–Smirnov statistics, summarised through `balance_table()`; `sensitivity_analysis()` reports the critical Rosenbaum Γ at which inference would no longer be significant (Rosenbaum, 2002). Blocked designs surface per-block diagnostics so imbalance is not hidden by averaging across strata.

Quickstart

The package ships with `hospital_staff`, a small example dataset for a treated/control workflow that runs without external data.

```
library(couplr)

data(hospital_staff)
treated <- transform(hospital_staff$nurses_extended,   id = nurse_id)
control <- transform(hospital_staff$controls_extended, id = nurse_id)

# Optimal one-to-one matching on three pre-treatment covariates, with
```

```r
# automatic scaling and solver selection.
m <- match_couples(
  left  = treated,
  right = control,
  vars  = c("age", "experience_years", "certification_level"),
  auto_scale = TRUE
)

# Balance on matching variables; analysis-ready paired output.
bal <- balance_diagnostics(m, treated, control,
  vars = c("age", "experience_years", "certification_level"))
balance_table(bal)
matched <- join_matched(m, treated, control)
```

On this dataset the matched pairs achieve absolute standardized differences below 0.15 on all three matching covariates. The same `match_couples()` call accepts `calipers`, `max_distance`, `block_id`, `method`, `strategy`, `replace`, and `ratio` for harder problems. Identification also requires conditional ignorability, positivity, and SUTVA; the matched output is designed to feed into an outcome regression for a doubly robust estimate (Stuart, 2010).

Comparison against established alternatives

We compare `couplr` against `MatchIt` and `optmatch`. The matching task is fixed: 1-to-1 optimal matching on a Mahalanobis distance with pooled within-group covariance, the convention `optmatch::match_on()` uses by default and `couplr` adopts as of 1.3.1. On the canonical Lalonde NSW dataset (Dehejia & Wahba, 1999; LaLonde, 1986) (185 treated, 429 control, eight covariates) all three packages produce identical absolute standardized mean differences to three decimal places (Figure 1b), with the remaining seven covariates at ≤ 0.128 (Stuart, 2010); per-covariate values are in `paper/lalonde-per-covariate.csv`.

Table 1 reports median wall-clock time on synthetic problems with the same eight-covariate structure, at six sizes from $n = 500$ to $n = 50{,}000$ with treated:control = 1:2. The margin widens with problem size: `couplr` is $9\times$ faster than `optmatch` at $n = 500$ and $24\times$ faster at $n = 20{,}000$, where it finishes in 11.4 seconds against 4.7 minutes, and $11\times$ to $24\times$ faster than `MatchIt` up to $n = 10{,}000$, the largest size `MatchIt` completes before `MatchIt::matchit(method = "optimal")` aborts inside the `optmatch` backend with an integer-size overflow. At $n = 50{,}000$ `couplr` completes in 79 seconds; both alternatives exceed the 300-second per-replicate cap. `couplr` reaches large dense problems through its dispatcher, which routes them to Jonker–Volgenant (Jonker & Volgenant, 1987). Under `memory_mode = "lazy"` the same $n = 50{,}000$ match completes in 61 seconds without ever building the matrix, returning the assignment the dense path returns (`paper/bench_scaling_lazy.R`).

Table 1: 1-to-1 optimal Mahalanobis matching, median wall-clock by problem size. Treated:control = 1:2; eight covariates; pooled within-group covariance; single core, single-threaded BLAS, on an Apple M4 Pro. Median of 5 / 5 / 3 / 3 / 1 / 1 replicates respectively for the rows; `optmatch_max_problem_size` set to `Inf` for $n \geq 10{,}000$. `couplr` is timed with `memory_mode = "dense"`, so all three packages materialize the distance matrix; the lazy path is reported in the text. `int overflow` marks an integer-size overflow (`result would exceed 2^31-1 bytes`) inside the `optmatch` backend reached through `MatchIt`; `timeout` marks a replicate exceeding the 300-second cap. Reproducible from `paper/bench_scaling.R` and `paper/bench_scaling_alternatives.R`.

Problem size ($n_t + n_c$)	`couplr`	`optmatch`	`MatchIt`
$167 + 333$	11 ms	99 ms	117 ms
$667 + 1{,}333$	144 ms	1.70 s	2.12 s
$1{,}667 + 3{,}333$	742 ms	12.9 s	15.7 s

Problem size ($n_t + n_c$)	couplr	optmatch	MatchIt
$3{,}333 + 6{,}667$	3.08 s	59.3 s	73.4 s
$6{,}667 + 13{,}333$	11.4 s	280 s	int overflow
$16{,}667 + 33{,}333$	78.5 s	timeout	timeout

Table 2 summarises feature coverage.

Table 2: Differentiating feature coverage; rows where all three packages support the feature (covariate distance, propensity-score matching, exact blocking, cost-matrix entry point) are omitted. *partial* indicates the feature is achievable but not via a first-class user-facing API.

Feature	couplr	MatchIt	optmatch
Per-variable calipers (named vector)	yes	yes	partial
Rectangular $n_t \neq n_c$, no padding	yes	partial	partial
Sparse-cost support	yes	no	yes
k-best assignments (Murty)	yes	no	no
Bottleneck (minimax) assignment	yes	no	no
Rosenbaum sensitivity bounds	yes	no	no

AI usage disclosure

The package was developed using a modern AI-assisted developer stack. The author designed the package architecture, made the algorithm-selection choices, and wrote the implementation in a TUI-first command-line workflow, using a self-hosted Qwen3-Coder-Next-REAP-48B-A3B model (mixture-of-experts coder, ~3B active per token, 4-bit MLX quantization, running on a single Apple M4 Pro) for code completion, refactoring, and boilerplate generation through Anthropic's Claude Code CLI, with requests routed to the local model and no cloud inference.

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